Techno-Economic Assessment and Energy Management of a Domestic Microgrid: A Comparative Study of Fixed vs. Dual-Axis PV Tracking with BESS

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Abstract

Residential photovoltaic systems often fall short of their full potential due to static panel orientations and unmanaged home energy consumption. Building upon an established academic research framework for dual-axis solar tracking optimization, this study expands this physical baseline into a complete, smart domestic microgrid. The system integrates a multi-physics solar model incorporating Faiman's thermal cell dynamics and GB 50797 inter-row shading constraints with an advanced Energy Management System (EMS). Operating at a 30-minute resolution, the EMS coordinates a Battery Energy Storage System (BESS) across real-world residential load profiles from a local utility provider database in Mulhouse, France (2024) to perform peak shaving and time-of-use (ToU) tariff arbitrage. Dynamic dual-axis tracking increases annual energy yield by +19.52% compared to fixed installation. When coupled with the BESS and EMS, the microgrid achieves a 60.2% self-sufficiency rate, significantly reducing annual grid purchases by 9.3%. Over a 20-year operational horizon, a Discounted Cash Flow (DCF) analysis reveals a Net Present Value (NPV) of €16,207.92 for the tracker configuration, while the fixed installation achieves a Levelized Cost of Energy (LCOE) of €0.1352/kWh, well below the average retail grid price of approximately €0.238/kWh. A sensitivity and scenario analysis further confirms the financial resilience of the investment under adverse market and regulatory assumptions. This work presents a comprehensive framework connecting solar tracking with home energy autonomy, bridging physical performance with long-term economic viability.

Keywords: Domestic Microgrid, Dual-Axis Solar Tracker, Energy Management System (EMS), Battery Energy Storage System (BESS), Faiman Thermal Model, Discounted Cash Flow (DCF)

1. Introduction

1.1. Background and motivation

The transition toward low-carbon residential energy infrastructure requires a fundamental shift from centralized power distribution toward localized, highly autonomous microgrids. The physical foundation and primary empirical dataset of this study stem from an academic research project initially carried out at Lycée Albert Schweitzer in Mulhouse, France [1]. This preliminary work focused on the mathematical and experimental optimization of a dual-axis solar tracking mechanism, establishing a baseline annual energy yield gain over static installations.

While individual solar tracking optimization provides valuable physical performance improvements, maximizing the full thermodynamic and financial benefits of residential renewable generation requires broader system-level integration. In real-world applications, solar production curves rarely match household demand profiles. Without intelligent energy storage and active dispatch strategies, high peak generation is frequently reduced into the main grid at low remuneration rates, while household electricity demand remains reliant on expensive grid supply during peak tariff hours. Driven by this challenge, the present study expands this physical solar baseline into a complete domestic microgrid framework, incorporating a Battery Energy Storage System (BESS) and an advanced Energy Management System (EMS) designed for real-world utility load environments.

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1.2. Literature review

1.2.1. Domestic microgrids and renewable potential in France

In France, the residential sector accounts for approximately 36% of national final electricity consumption, presenting a major target for localized decarbonization and grid flexibility initiatives [2]. The national energy framework, supported by the PPE (*Programmation Pluriannuelle de l'Énergie*) and recent regulatory incentives for self-consumption (*autoconsommation individuelle et collective*), has accelerated the adoption of rooftop photovoltaic (PV) systems. However, static residential setups in northern and eastern regions, such as Grand Est (Mulhouse), face substantial seasonal solar variance, where winter daily irradiance can drop below 1.5 kWh/m^2/day compared to summer peaks exceeding 6.5 kWh/m^2/day [3].

To mitigate these resource fluctuations, domestic microgrids combining PV, BESS, and Time-of-Use (TOU) tariff optimization have emerged as a pivotal research area. Recent field studies across French distribution networks indicate that pairing residential solar arrays with dynamic energy management can increase local self-consumption rates from an average of 30–35% up to 60–75%, while relieving peak transformer stress during the critical 18:00–20:00 evening grid peak [4, 5].

1.2.2. Global perspectives on residential microgrids

On a global scale, the deployment of residential microgrids is advancing rapidly across Europe, North America, and Asia-Pacific, driven by rising utility electricity tariffs and falling battery capital costs [6, 7]. In Germany and California, over 70% of new residential PV installations are equipped with integrated lithium-ion storage systems to hedge against retail price volatility and grid instability [8].

International research has extensively focused on optimizing energy dispatch strategies across diverse operational constraints. Studies by various authors demonstrate that intelligent EMS algorithms operating on high-resolution load datasets (e.g., 15-minute to 30-minute intervals) are essential for achieving optimal Time-of-Use (TOU) arbitrage and dynamic peak shaving [9, 10]. Furthermore, integrating multi-physics solar performance, accounting for temperature-dependent cell efficiency losses and inter-row geometric shading constraints, into long-term financial Discounted Cash Flow (DCF) models remains a key requirement for evaluating true project viability across a 20-year operational lifecycle [11, 12].

2. System Modeling and Architecture

2.1. Physical system description and microgrid architecture

The microgrid framework is evaluated on a real-world residential single-family house (R+1) located in Mulhouse, France (47.75°N, 7.34°E). The building features a total living floor area of approximately 120 m^2, serving a typical residential household. The structural footprint includes an unshaded roof surface optimized for solar orientation, providing sufficient mechanical and physical area for photovoltaic deployment.

The microgrid framework is evaluated on a residential single-family house (R+1 two-story architecture, 120 m^2 living area) located in Mulhouse, France (47.75°N, 7.34°E). The system consists of three main components:

- **Photovoltaic Array:** A 7.6 kWp PV installation covering a total surface area of 36 m^2, mounted on a dual-axis tracking system to maximize solar energy capture (see Table 1).
- **Battery Energy Storage System (BESS):** A Huawei LUNA2000-10-S0 system providing 10 kWh of nominal capacity and a 8 kW hybrid inverter, utilizing lithium iron phosphate ($LiFePO_4$) chemistry (see Table 2).
- **Grid Interconnection Point (GIP):** A three-phase low-voltage grid connection (400 V/230 V, 50 Hz) managed with a 30-minute sampling resolution under French Time-of-Use (TOU) tariffs.

Table 1: Electrical and thermal characteristics of the photovoltaic module (STC conditions: 1000 W/m^2, 25°C, AM = 1.5)

Technical Parameter	**Symbol**	**Value**
Module surface area	S	1.80 m^2
Peak power	P_{STC}	380 Wp
Voltage at max power point	V_{mp}	42.8 V
Current at max power point	I_{mp}	11.68 A
Open-circuit voltage	V_{oc}	51.2 V
Short-circuit current	I_{sc}	12.52 A
Temperature coefficient of power	γ	−0.35%/°C
Nominal operating cell temperature	$NOCT$	45°C

2.2. Geometric sizing and inter-row spacing

The deployment of a tracker-based photovoltaic array on a flat roof imposes a strict spacing constraint to prevent mutual shading between successive rows.

In accordance with international solar plant design standards (such as GB 50797), the rear panel must remain fully unshaded by the front panel between 09:00

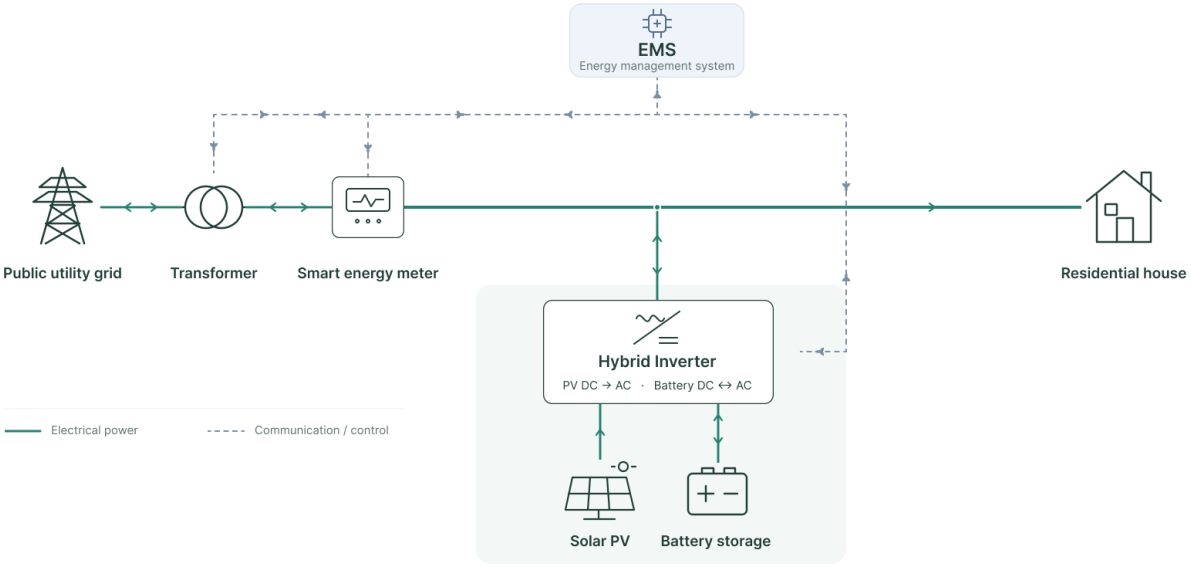


Figure 1: Schematic diagram of the residential microgrid architecture, illustrating power flows (AC/DC) and EMS communication links.

Table 2: Technical specifications of the reference BESS (Huawei LUNA2000-10-S0)

Property / Characteristic	Symbol	Value	Unit
Reference model	—	Huawei LUNA2000-10-S0	—
Cell chemistry	—	$LiFePO_4$	—
Nominal energy capacity	E_{nom}	10	kWh
Maximum charge power	$P_{ch,max}$	5	kW
Maximum discharge power	$P_{dis,max}$	5	kW
Charge efficiency	η_{ch}	95	%
Discharge efficiency	η_{dis}	95	%
Round-trip efficiency	η_{rt}	90.25	%
Lower critical threshold	SoC_{min}	10	%
Upper saturation threshold	SoC_{max}	100	%
Maximum depth of discharge	DoD_{max}	90	%
Nominal voltage range	V_{nom}	360 – 400	V

and 15:00 solar time on the winter solstice. Although no single textual formula is fixed by French regulation for this specific case, this analytical approach is consistent with European engineering practice.

2.2.1. Analytical spacing model

The minimum distance D between the base of two successive rows is expressed as:

$$D = L\cos\beta + L\sin\beta\frac{0.707\tan\phi + 0.4338}{0.707 - 0.4338\tan\phi} \quad (1)$$

where D is the minimum inter-row distance (m), L the module length (m), β the module tilt (deg), and ϕ the local latitude (deg).

For Mulhouse ($\phi = 47.75°$), standard modules ($L = 1.80$ m), and a critical winter tilt $\beta_{max} = 45°$, Eq. (1) yields a minimum center-to-center spacing of $D \approx 8.00$ m, which satisfies the winter insolation criterion.

2.2.2. Deduction of the usable active area (A_{pv})

For a flat-roof footprint of 125 m^2 (12.5 m North-South × 10 m East-West):

- **North-South layout:** A pitch $D = 8.00$ m allows exactly two complete rows within the available 12.5 m.
- **East-West layout:** The 10 m width accommodates 10 modules per row.

The layout yields 20 modules, giving a validated active area:

$$A_{pv} = 20 \times 1.80\ \text{m}^2 = 36.0\ \text{m}^2 \quad (2)$$

This area provides a realistic installed capacity while avoiding any mutual winter shading losses.

2.3. Photovoltaic electrical and thermal modeling

The electrical performance of the photovoltaic module is characterized by its Fill Factor (FF), nominal efficiency under Standard Test Conditions (STC, $G_{STC} = 1000$ W/m^2, $T_{STC} = 25°$C), and a detailed breakdown of losses via the Performance Ratio ($PR(t)$).

2.3.1. Fill Factor and Nominal Efficiency

The Fill Factor (FF) defines the squareness of the current-voltage characteristic curve [13]:

$$FF = \frac{P_{mp}}{V_{oc} \cdot I_{sc}} \approx 0.780 \tag{3}$$

Consequently, the nominal peak power of each module is expressed as $P_{mp,\mathrm{STC}} = V_{oc} \cdot I_{sc} \cdot FF = 500$ Wc, yielding a total peak power of 10 kWp for the 36 m^2 array (comprising 20 modules). The instantaneous and nominal electrical efficiencies are given by:

$$\eta(t) = \frac{P_{mp}(t)}{A_{pv} \cdot G_{\mathrm{inc}}(t)}, \quad \eta_{\mathrm{STC}} = \frac{P_{mp,\mathrm{STC}}}{A_{pv} \cdot G_{\mathrm{STC}}} \approx 27.78\% \tag{4}$$

2.3.2. Performance Ratio Decomposition

The instantaneous overall electrical efficiency $\eta(t)$ is related to η_{STC} through the Performance Ratio $PR(t)$ [14]:

$$\eta(t) = \eta_{\mathrm{STC}} \cdot PR(t) \tag{5}$$

The $PR(t)$ factor aggregates thermal and fixed loss components [15]:

$$PR(t) = f_{\mathrm{temp}}(t) \cdot f_{\mathrm{inv}} \cdot f_{\mathrm{cables}} \cdot f_{\mathrm{opt}} \cdot f_{\mathrm{soiling}} \tag{6}$$

Fixed loss factors account for inverter conversion ($f_{\mathrm{inv}} = 0.970$), ohmic wiring losses ($f_{\mathrm{cables}} = 0.985$), optical/IAM losses ($f_{\mathrm{opt}} = 0.975$), and soiling ($f_{\mathrm{soiling}} = 0.980$), resulting in a combined fixed efficiency factor $f_{\mathrm{fix}} \approx 0.912$.

2.3.3. Thermal Modeling and Faiman Law

The thermal factor $f_{\mathrm{temp}}(t)$ captures efficiency degradation due to cell heating:

$$f_{\mathrm{temp}}(t) = 1 + \gamma \cdot (T_{\mathrm{cell}}(t) - T_{\mathrm{STC}}) \tag{7}$$

where $\gamma \approx -0.35\%/^\circ$C is the temperature coefficient. The cell operating temperature $T_{\mathrm{cell}}(t)$ is determined using the empirical Faiman thermal model [16, 17, 18]:

$$T_{\mathrm{cell}}(t) = T_{\mathrm{amb}}(t) + \frac{G_{\mathrm{inc}}(t)}{U_0' + U_1' \cdot WS(t)} \tag{8}$$

where $T_{\mathrm{amb}}(t)$ is the ambient temperature, $WS(t)$ is the wind speed, and $U_0' \approx 25.0$ W/m^{2}K and $U_1' \approx 6.84$ W/m^{2}K/(m/s) represent the apparent heat transfer coefficients (implicitly incorporating the effective optical-to-thermal conversion term $\alpha - \eta_{\mathrm{STC}}$).

2.3.4. Meteorological dataset and solar irradiance modeling

To evaluate the dynamic electrical and thermal models established previously, the simulation framework relies on a high-resolution real-world meteorological and radiometric dataset specifically processed for the location of Mulhouse, France (47.75°N, 7.34°E) [1].

The database comprises 17,568 timestamps recorded at a 30-minute sampling resolution covering the entire year of 2024. The primary variables extracted from the dataset to feed the photovoltaic and thermal models include:

- **Solar Irradiance Components:** direct, diffuse, and reflected irradiance components projected onto the dual-axis tracking plane [1], which sum up to yield the total incident irradiance $G_{\mathrm{inc}}(t)$.
- **Ambient Temperature ($T_{\mathbf{2m}}$):** Temperature measurements recorded at 2 meters height, serving as the baseline ambient temperature $T_{\mathrm{amb}}(t)$ for the Faiman thermal balance.
- **Wind Velocity ($WS_{\mathbf{10m}}$):** Wind speed measurements at 10 meters height, utilized to quantify forced convective heat dissipation across the photovoltaic modules.

2.4. Energetic Performance and PV System Efficiency

A comparative analysis was carried out to evaluate the performance of a 36 m^2 photovoltaic array (20 modules of 1.8 m^2, $\eta_{\mathrm{STC}} = 21.11$ %, $P_p \approx 7.60$ kWc) operating under the two configurations. Table 3 reports the annual balance of the incident solar energy (E_{inc}) and of the net electrical energy produced (E_{elec}).

The annual average electrical conversion efficiency η_{annual} reflects the global effectiveness of converting the solar resource into useful electricity over the full year:

$$\eta_{\mathrm{annual}} = \frac{E_{\mathrm{elec}}}{E_{\mathrm{inc}}} \tag{9}$$

Applying this definition to the annual data yields $\eta_{\mathrm{annual,fixed}} = 18.73$ % and $\eta_{\mathrm{annual,tracker}} = 18.67$ %. Given their near-identity, a single effective PV efficiency $\eta \approx 18.7$ % is retained for both configurations in the subsequent energy management and economic analyses.

The Performance Ratio (PR) isolates system losses by comparing this effective efficiency to the manufacturer's nominal value under Standard Test Conditions ($\eta_{\mathrm{STC}} = 21.11$ %):

$$PR = \frac{\eta_{\mathrm{annual}}}{\eta_{\mathrm{STC}}} \tag{10}$$

Table 3: Comparative annual energy balance ($S = 36\,\mathrm{m}^2$) [1].

Physical quantity	Fixed	Tracker	Gain
Incident solar energy (E_{inc})	43.01 MWh/yr	51.55 MWh/yr	+19.85 %
Produced electrical energy (E_{elec})	8.05 MWh/yr	9.63 MWh/yr	+19.52 %
Additional electrical production	–	–	+1 572.07 kWh/yr

yielding $PR_{\mathrm{fixed}} = 88.70\,\%$ and $PR_{\mathrm{tracker}} = 88.46\,\%$.

The gap between the nominal and effective efficiencies (21.11 % → 18.7 %) originates from the STC reference at a 25 °C junction temperature: under real operating conditions, cell heating under irradiation and ambient temperature lowers the module voltage, reducing the annual average efficiency by about 2.4 points. The slightly lower *PR* of the tracker (−0.24 %) despite its +19.52 % gain in energy production is explained by its near-zero incidence angle, which raises the average cell temperature T_{cell} according to the Faiman thermal model and induces an additional thermal loss governed by $\gamma = -0.35\,\%/^{\circ}\mathrm{C}$.

3. Residential Load Modeling

3.1. Residential load profile hypothesis (Mulhouse, 2024)

To assess the coverage rate and the synergy of the $36\,\mathrm{m}^2$ photovoltaic array, an electrical demand model is defined for a representative single-family house (two-storey R+1, $120\,\mathrm{m}^2$ living area, four occupants). The building is assumed recent and energy-efficient (RE2020/RT2012 standard), located in the Mulhouse region (climate zone H1b, semi-continental, DJU $\approx 2\,650\,\mathrm{K}\cdot\mathrm{d}$) [19].

The annual electricity consumption for the 2024 reference year is set to:

$$E_{\mathrm{cons,\,annual}} = 9\,000\,\mathrm{kWh/yr} \tag{11}$$

This value aggregates the following end-use breakdown:

1. **Thermodynamic heating** (3 900 **kWh/yr**)**:** an air-to-water heat pump (COP ≈ 3.1) [20] covers the building thermal demand $Q_{\mathrm{th}} \approx 12\,000\,\mathrm{kWh_{th}/yr}$, giving $E_{\mathrm{heat}} = Q_{\mathrm{th}}/\mathrm{COP} \approx 3\,870\,\mathrm{kWh}$.
2. **Domestic hot water** (1 200 **kWh/yr**)**:** thermodynamic water heater (COP ≈ 2.7), 200 L/day at 55 °C [21].
3. **Specific electricity** (2 800 **kWh/yr**)**:** appliances (class A+++), induction cooking, LED lighting and multimedia [22, 23].
4. **Summer cooling** (700 **kWh/yr**)**:** reversible heat pump operation during 2024 summer sequences (SEER ≈ 4.0) [24].
5. **Auxiliaries and ventilation** (400 **kWh/yr**)**:** hygro-adjustable mechanical ventilation (VMC) and hydraulic circulators.

yielding:

$$E_{\mathrm{cons,\,total}} = E_{\mathrm{heat}} + E_{\mathrm{DHW}} + E_{\mathrm{specific}} + E_{\mathrm{cool}} + E_{\mathrm{aux}} \tag{12}$$

This 9 000 kWh/yr framework is consistent with a fully electrified dwelling in zone H1b, and allows a direct comparison against the simulated PV production (fixed: 8 055 kWh/yr; tracker: 9 627 kWh/yr) with a load profile that integrates the cross-seasonality of winter heating and summer cooling.

3.2. Data source and temporal scope

The electrical demand is reconstructed from the public *Enedis* database [25], the main French distribution system operator. The dataset covers the complete 2024 calendar year, from 1 January 2024 00:00 to 31 December 2024 23:45, at a 30-minute resolution (17 568 samples). These national usage profiles allow Enedis to continuously estimate the consumption of delivery points (PDL) not equipped with real-time smart meters, ensuring supply–demand balance at the national scale [25]. Their half-hourly resolution matches the solar simulation time step exactly, enabling a coherent net power balance at each step.

3.3. Profile nomenclature and coefficient structure

The database is split into consumer categories (residential and business) and further into sub-profiles [27]:

- **Subscribed power** (≤ 6 **kVA vs** > 6 **kVA**)**:** profiles `RES1`/`RES11` correspond to small-to-medium dwellings with a subscribed power ≤ 6 kVA; profiles `RES2`/`RES22` cover dwellings above 6 kVA (9, 12 kVA or more).
- **Weekday and weekend separation** (`_SEM` **vs** `_WE`)**:** `RES11WE_SEM` holds coefficients for working days (Monday–Friday, excluding holidays), and is strictly zero on weekends and public holidays; conversely `RES11WE_WE` holds Saturday, Sunday and holiday coefficients, and is zero on working days.

- **Prepared vs adjusted coefficients:** `COEFFICIENT_PREPARE` is the base statistical calendar shape (average hourly habits of a French household); `COEFFICIENT_AJUSTE` corrects it dynamically with observed outdoor temperature, capturing the dwelling thermosensitivity [25].

3.4. Load profile reconstruction

Rather than using fixed power time series (kW) or absolute energy records (kWh), the analysis exploits the normalized Enedis coefficients. Each 30-minute step is mapped to the average power drawn by the household:

$$P(t) = C_{\text{adjusted}}(t) \times \mathcal{F}_{\text{scale}} \tag{13}$$

where $\mathcal{F}_{\text{scale}}$ is a proportionality factor set so that the annual integral of $P(t)$ matches the target annual consumption:

$$\sum_{t=1}^{N_t} P(t)\,\Delta t = E_{\text{cons, annual}} = 9\,000\,\text{kWh}. \tag{14}$$

The adjusted coefficient is retained throughout, since it carries the weather-driven heating and cooling calls essential to the Mulhouse case study.

3.5. Computational implementation

The Enedis coefficients are processed through a Python pipeline based on `pandas`. Raw timestamps are resampled onto a 30-minute grid, the weekday (`RES11WE_SEM`) and weekend (`RES11WE_WE`) subprofiles are merged on that common grid and summed to reconstruct the global `RES11WE` coefficient, and the resulting time series is scaled by the factor $\mathcal{F}_{\text{scale}}$ of Eq. (13). The resulting scale factor is $\mathcal{F}_{\text{scale}} = 0.5158$ kW, corresponding to an average load of approximately 1.03 kW per half-hour step over the full year. The output provides the instantaneous load power $P_{\text{load}}(t)$ at each of the 17 568 time steps, ready for the EMS simulation.

4. Energy Management System and BESS Integration

The final functional block of the microgrid integrates a Battery Energy Storage System (BESS) driven by an Energy Management System (EMS) [31]. Its role is to absorb PV fluctuations, maximize self-consumption, smooth the power draw on the distribution grid, and support demand-response requirements [33, 34].

4.1. Power balance and sign conventions

At each discrete time step t ($\Delta t = 30$ min in the datasets), the instantaneous power balance within the microgrid reads [35]:

$$P_{\text{load}}(t) = P_{\text{gen}}(t) + P_{\text{batt}}(t) + P_{\text{grid}}(t) \tag{15}$$

with the following sign conventions:

- $P_{\text{load}}(t) \geq 0$: active power consumed by the site load;
- $P_{\text{gen}}(t) \geq 0$: active power injected by the PV generator;
- $P_{\text{batt}}(t)$: power at the BESS terminals ($P_{\text{batt}} > 0$ discharging, $P_{\text{batt}} < 0$ charging);
- $P_{\text{grid}}(t)$: power exchanged with the distribution grid ($P_{\text{grid}} > 0$ import, $P_{\text{grid}} < 0$ export).

4.2. State of Charge (SoC) modeling

The storage dynamics are governed by the state of charge $SoC(t)$. Accounting for the charge and discharge efficiencies η_{ch} and η_{dis}, the discrete-time state equation is [28]:

$$SoC(t + \Delta t) = SoC(t) + \left(\frac{\eta_{ch}\,P_{ch}(t) - P_{dis}(t)/\eta_{dis}}{E_{\text{nom}}}\right)\Delta t \tag{16}$$

where E_{nom} is the nominal BESS capacity (Wh). The charge and discharge powers P_{ch} and P_{dis} are bounded by the maximum admissible power $P_{max,batt}$ imposed by the inverter and cell chemistry [28].

4.3. EMS decision algorithm

At each time step, the EMS evaluates the unconstrained net power $P_{\text{net}}(t) = P_{\text{gen}}(t) - P_{\text{load}}(t)$ and applies a sequential decision logic structured around three main cases [31, 32]:

1. **Case 1: Production surplus ($P_{\text{net}}(t) > 0$).**
   - *If $SoC(t) < SoC_{max}$ (100%):* the surplus is stored first. The charging power is:

$$P_{\text{batt}}(t) = -\min\left(P_{\text{net}}(t),\ P_{ch,max},\ \frac{(SoC_{max} - SoC(t))\,E_{\text{nom}}}{\eta_{ch}\,\Delta t}\right) \tag{17}$$

   the remainder being exported:
   $P_{\text{grid}}(t) = -(P_{\text{net}}(t) + P_{\text{batt}}(t))$.

- *If $SoC(t) = 100\%$:* the storage is saturated, the battery is idle ($P_{\text{batt}}(t) = 0$), and the full surplus is exported: $P_{\text{grid}}(t) = -P_{\text{net}}(t)$.

2. **Case 2: Production deficit during peak hours.**
   - *If $SoC(t) > SoC_{min}$ (critical threshold at 10%):* the EMS discharges the battery to cover the load:

$$P_{\text{batt}}(t) = \min\Big(|P_{\text{net}}(t)|,\ P_{dis,max},\ \frac{(SoC(t) - SoC_{min})\, E_{\text{nom}}\, \eta_{dis}}{\Delta t}\Big) \tag{18}$$

     the missing energy being drawn from the grid: $P_{\text{grid}}(t) = |P_{\text{net}}(t)| - P_{\text{batt}}(t)$.
   - *If $SoC(t) \leq SoC_{min}$:* discharge is inhibited ($P_{\text{batt}}(t) = 0$) to avoid irreversible cell degradation; the grid supplies the full deficit: $P_{\text{grid}}(t) = |P_{\text{net}}(t)|$.
3. **Case 3: Off-peak hours.**
   - During low-demand, low-tariff periods, the EMS may trigger anticipatory BESS charging from the grid to reach an optimal SoC before the next peak [29].

4.4. Advanced off-peak management (Case 3)

4.4.1. Three objectives of off-peak management

1. **Financial arbitrage:** purchase electricity at the minimum marginal cost $C_{\text{grid}}(t)$ (off-peak tariff) to avoid the peak tariff [30].
2. **Peak shaving:** guarantee sufficient stored energy before the peak so that the BESS covers the fraction of demand exceeding the contracted grid power $P_{\text{grid,contracted}}$, avoiding penalties [32].
3. **Predictive anticipation (SoC_{target}):** a systematic charge to 100% at night would leave no free capacity to absorb a strong PV surplus the next morning. The target SoC_{target} must therefore be computed dynamically from the production and load forecasts ($\hat{P}_{\text{gen}}$, $\hat{P}_{\text{load}}$) [34].

4.4.2. Mathematical formulation

The binary tariff indicator is defined as:

$$I_{\text{offpeak}}(t) = \begin{cases} 1 & \text{if } t \in [\text{off-peak hours}] \\ 0 & \text{otherwise} \end{cases} \tag{19}$$

The target level at the end of the off-peak period integrates the forecasted PV surplus of the next day ($\Delta E_{\text{PV,surplus}}$):

$$SoC_{\text{target}}(t) = \min\left(SoC_{max},\ SoC_{max} - \frac{\Delta E_{\text{PV,surplus}}}{E_{\text{nom}}}\right) \tag{20}$$

with:

$$\Delta E_{\text{PV,surplus}} = \int_{t_{\text{morning}}}^{t_{\text{peak}}} \max\Big(0,\ \hat{P}_{\text{gen}}(\tau) - \hat{P}_{\text{load}}(\tau)\Big)\, d\tau \tag{21}$$

If no surplus is forecast, $SoC_{\text{target}} = 100\%$; if a strong surplus is expected, a free margin is kept to absorb it at no cost.

When $I_{\text{offpeak}}(t) = 1$ and $SoC(t) < SoC_{\text{target}}(t)$, the EMS computes the required charging power:

$$P_{\text{batt}}(t) = -\min\Big(P_{ch,max},\ \frac{(SoC_{\text{target}}(t) - SoC(t))\, E_{\text{nom}}}{\eta_{ch}\, \Delta t},\ P_{\text{grid,max_import}} - P_{\text{load}}(t) + P_{\text{gen}}(t)\Big) \tag{22}$$

The three arguments of the min() are: (i) the inverter/battery hardware limit $P_{ch,max}$; (ii) the exact power needed to reach SoC_{target} within Δt; (iii) a guard against exceeding the subscribed power $P_{\text{grid,max_import}}$ while simultaneously supplying P_{load} and charging the battery [35].

5. Financial and Economic Analysis of the Domestic Microgrid

The economic evaluation of the photovoltaic microgrid coupled with the battery storage system (PV–BESS) relies on an integrated dynamic model. This approach combines the high-resolution physical energy behaviour ($\Delta t = 0.5$ h) with financial discounting mechanisms over the system lifetime ($N = 20$ years).

5.1. Study framework, input data and assumptions

The financial profitability is assessed from the simulation of a residential load profile of type `RES11WE` (single-family house with sustained weekend occupancy), coupled with the real meteorological data recorded in Mulhouse during 2024. The system includes a Huawei LUNA2000 BESS operated within the state-of-charge range SoC $\in [10\%, 100\%]$ ($\text{DoD}_{\text{max}} = 90\%$) [28].

5.1.1. Grid tariff framework

To reflect the French tariff structure, the system is subject to an off-peak/peak (Heures Pleines / Heures Creuses, HP/HC) supply contract based on the EDF Tarif Bleu. The non-self-consumed and non-stored production surplus is valorised through the Obligation d'Achat (EDF OA) contract at a fixed injection tariff [36].

5.2. Dynamic modeling of intraday financial flows

At each time step t ($\Delta t = 0.5$ h, for $t = 1, \ldots, 17\,568$ corresponding to the 8 784 h of a leap year), the net power injected into or drawn from the grid $P_{\text{grid}}(t)$ (in kW) generates an instantaneous financial flow $\phi(t)$ (in €):

$$\phi(t) = \begin{cases} -P_{\text{grid}}(t)\Delta t\, C_{\text{tarif}}(t), & \text{if } P_{\text{grid}}(t) > 0 \\ |P_{\text{grid}}(t)|\Delta t\, C_{\text{inj}}, & \text{if } P_{\text{grid}}(t) < 0 \end{cases} \tag{23}$$

where the purchase tariff $C_{\text{tarif}}(t)$ varies dynamically according to the hourly register:

$$C_{\text{tarif}}(t) = \begin{cases} C_{\text{HC}}, & \text{if Hour}(t) \in \{22, 23, 0, 1, 2, 3, 4, 5\} \\ C_{\text{HP}}, & \text{otherwise.} \end{cases} \tag{24}$$

The cumulative financial balance (Net Portfolio) at day $D \in [1, 365]$ is computed by discrete temporal integration of the instantaneous flows :

$$\text{Net Portfolio}(D) = \sum_{t=1}^{N_D} \phi(t) - \frac{D}{365} C_{\text{fixe}} \tag{25}$$

where $N_D = D \times 48$ is the cumulative number of time steps at day D.

At the end of the first year ($y = 1$), the reference gross bill $B_{\text{ref},1}$ (without PV–BESS) and the optimised bill $B_{\text{opt},1}$ are evaluated to extract the initial gross saving S_1:

$$S_1 = B_{\text{ref},1} - B_{\text{opt},1} \tag{26}$$

5.3. Investment and operating cost structure (CAPEX / OPEX)

The economic analysis includes all life-cycle costs [37]:

1. **CAPEX (I_0):** initial investment covering the PV modules, the mounting structure (fixed or tracker), the hybrid inverter, the Huawei LUNA2000 battery, the IoT acquisition gateway (ESP32) and installation.
2. **Recurring annual OPEX (M_{rec}):** operation, insurance and preventive maintenance costs, set at 1% of the initial CAPEX per year:

$$M_{\text{rec}} = 0.01\, I_0 \tag{27}$$

3. **Exceptional reinvestment CAPEX (M_{extra}):** a one-off expense at year $y = 10$, corresponding to the preventive replacement of the hybrid inverter or the refurbishment of the BESS power components [38].

The total operating cost of year y is therefore:

$$M_y = \begin{cases} M_{rec} + M_{extra}, & \text{if } y = 10 \\ M_{rec}, & \text{otherwise.} \end{cases} \tag{28}$$

5.4. Multi-year projection and financial discounting

To project the performance over $N = 20$ years, two crossed temporal dynamics are modelled:

5.4.1. Electricity price escalation

The grid tariff undergoes an estimated annual inflation of $r_e = 3.0\%$. The gross saving realised in year y follows the geometric law:

$$S_y = S_1\,(1 + r_e)^{y-1} \tag{29}$$

5.4.2. Physico-chemical degradation of production

The annual PV production $E_{\text{pv},y}$ decreases at the linear degradation rate of $\delta = 0.5\%/\text{yr}$ of the modules [39]:

$$E_{\text{pv},y} = E_{\text{pv},1}\,(1 - \delta)^{y-1} \tag{30}$$

5.4.3. Discounted net cash flow ($NCF_{actu,y}$)

The annual net cash flow ($\text{NCF}_y = S_y - M_y$) is brought back to present value at year t_0 using the discount rate $d = 4.0\%$:

$$\text{NCF}_{\text{actu},y} = \frac{S_y - M_y}{(1 + d)^y} \tag{31}$$

5.5. Economic performance indicators (KPIs)

5.5.1. Net Present Value (NPV)

The NPV measures the direct financial value creation generated by the microgrid implementation beyond the simple repayment of the invested capital [40]:

$$\text{NPV} = -I_0 + \sum_{y=1}^{N} \frac{S_y - M_y}{(1 + d)^y} \quad [€] \tag{32}$$

A project is considered financially viable and profitable if NPV > 0.

Table 4: Reference tariff and economic parameters (t_0 = 2024)

Economic / Tariff Parameter	Symbol	Value	Unit
Grid purchase price — Peak hours (06h–22h)	C_{HP}	0.2700	€/kWh
Grid purchase price — Off-peak hours (22h–06h)	C_{HC}	0.2068	€/kWh
Surplus injection tariff (EDF OA)	C_{inj}	0.1301	€/kWh
Fixed annual grid subscription (6–9 kVA)	C_{fixe}	170.00	€/yr
Annual electricity inflation rate	r_e	3.0	%
Financial discount rate	d	4.0	%
Annual PV degradation rate	δ	0.5	%/yr
Project evaluation period	N	20	years

5.5.2. *Internal Rate of Return (IRR)*

The IRR represents the discount rate d^* that exactly cancels the NPV of the project [40, 41]:

$$\text{NPV}(d^*) = -I_0 + \sum_{y=1}^{N} \frac{S_y - M_y}{(1+d^*)^y} = 0 \implies \text{IRR} = d^* \quad [\%] \tag{33}$$

The project is attractive when IRR > d.

5.5.3. *Discounted Payback Period (PBP)*

The PBP designates the minimum duration y^* (in years) required for the cumulative discounted cash flows to offset the initial investment I_0 [40, 41]:

$$PBP = \min\left\{ y^* \;\middle|\; \sum_{y=1}^{y^*} \frac{S_1\,(1+r_e)^{y-1} - M_y}{(1+d)^y} \geq I_0 \right\} \quad \text{[years]} \tag{34}$$

5.5.4. *Levelised Cost of Energy (LCOE)*

The LCOE represents the full discounted cost of each kWh produced by the PV system over its entire life cycle. It is obtained by dividing the discounted sum of expenses (CAPEX + OPEX) by the discounted sum of generated energy [37]:

$$LCOE = \frac{I_0 + \sum_{y=1}^{N} \frac{M_y}{(1+d)^y}}{\sum_{y=1}^{N} \frac{E_{\text{pv},1}\,(1-\delta)^{y-1}}{(1+d)^y}} \quad \text{[€/kWh]} \tag{35}$$

The objective is always to minimise the LCOE; in this study, the analysis is considered successful and the project worth investing in if the computed LCOE value is lower than the grid purchase price, whether during off-peak or peak hours [37, 40].

6. Results and Analysis

This section presents the results of the annual microgrid simulation. The analysis focuses on the operational dynamics of the Battery Energy Storage System (BESS) and the decision-making logic of the Energy Management System (EMS), culminating in a comprehensive 20-year Discounted Cash Flow (DCF) financial evaluation to assess the long-term economic viability of the project.

6.1. *Technical Analysis: Comparative Energetic Performance and EMS/BESS Dynamics*

This section evaluates the technical performance of the proposed microgrid architecture, focusing on the physical energy flows and the operational efficiency of the Energy Management System (EMS). The analysis is structured in three progressive steps. First, a macroscopic comparative assessment is conducted to quantify the annual energetic balance and the overall impact of the dual-axis tracker compared to a fixed installation. Second, the seasonal dynamics are investigated through representative weekly samples (summer vs winter) to highlight the system's response to extreme grid-dependency scenarios. Finally, the strategic effectiveness of the EMS is assessed through averaged daily profiles, demonstrating how the Battery Energy Storage System (BESS) implements advanced peak shaving and window shifting to maximize self-consumption and reduce operational costs.

6.1.1. *Comparative Energetic Balance: Fixed vs Dual-Axis Tracker*

The simulation extracted the annual energy balance based on the 17,568 half-hour intervals of the year 2024 in Mulhouse, France. Table 5 presents the comparative synthesis of the photovoltaic system's performance coupled with the 10 kWh BESS.

The integration of the dual-axis solar tracker increases the annual production by +19.5% compared to a fixed installation, rising from 8,054.7 kWh to 9,626.8 kWh. This increase in solar capture has a direct impact on the energy flows within the microgrid.

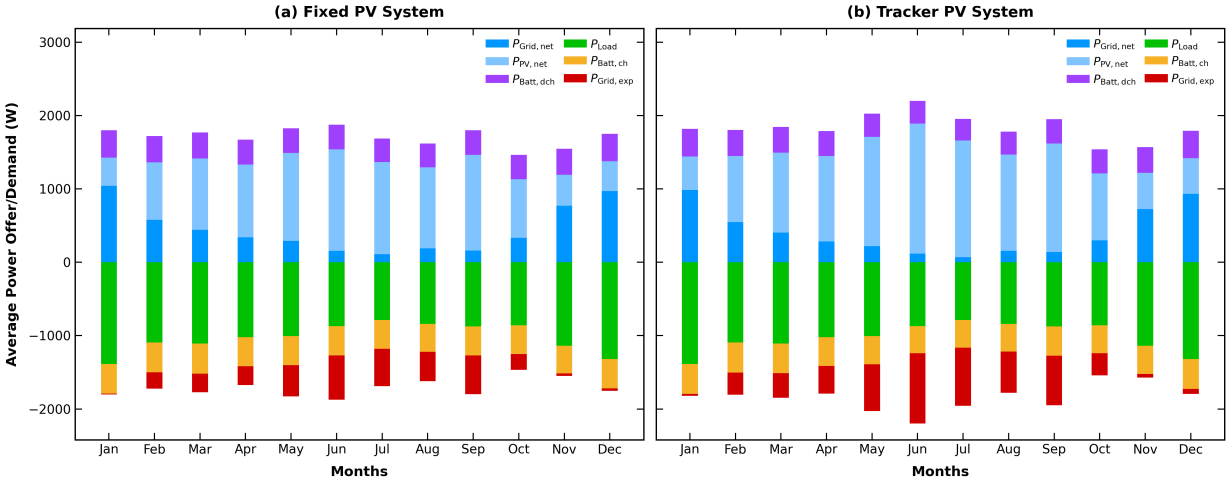


Figure 2: Monthly average power balance comparison between the (a) Fixed PV System and (b) Tracker PV System.

Table 5: Comparative annual energy balance of the microgrid (Fixed vs. Tracker)

Energetic Parameter	Fixed	Tracker	Gap
Solar Production (E_{gen})	8,054.7 kWh	9,626.8 kWh	+19.5%
Grid Purchases (E_{pur})	3,946.2 kWh	3,578.9 kWh	−9.3%
Grid Sales (E_{sale})	2,548.7 kWh	3,721.6 kWh	+46.0%
Self-Sufficiency (Tx_{auto})	56.2%	60.2%	+4.1 pts

On the one hand, the surplus energy injected into the grid increases significantly +46.0%, reaching 3,721.6 kWh for the tracker. On the other hand, the energy consumption drawn from the grid (purchases) decreases by -9.3% (from 3,946.2 kWh to 3,578.9 kWh). Consequently, the household's self-sufficiency rate increases from 56.2% (fixed) to 60.2% (tracker), demonstrating the synergy between dynamic solar tracking and battery storage in reducing grid dependency.

To visualize this energetic balance, Figure 2 presents the monthly average power offer and demand for both configurations. The stacked bar charts clearly show that the tracker configuration yields a visibly higher PV contribution throughout the spring and summer months. This increased generation leads to a substantial reduction in grid imports compared to the fixed system. However, during winter months (November to February), both configurations show a dominant reliance on grid imports to meet heating loads, confirming that the tracker's benefit is primarily seasonal.

6.1.2. *Seasonal Impact on Grid Dependency and System Performance*

The evolution of the SoC over representative seasonal samples provides insight into the system's dynamics. Figure 3 and Figure 4 present the extracted samples for a summer week (July 1-7) and a winter week (January 10-17), respectively.

To observe the extreme operational limits of the system, Figure 3 presents a detailed daily sample from a summer week (July 1-7).

The top panel illustrates that the PV generation drastically overwhelms the load, with the tracker system producing peak powers exceeding 6,500 W. Because the household load remains relatively low and flat (500 to 1,000 W), a massive energy surplus is generated. The middle panel shows the EMS response: the BESS heavily charges to absorb this surplus, hitting the maximum charging limit of -5,000 W for both configurations. The battery power profile exhibits sharp, square-wave-like charging spikes during the morning peaks, constrained by the inverter's maximum rated power. The bottom panel (SoC) clearly demonstrates the battery's saturation behavior. The BESS cycles between the minimum threshold (10%) and maximum capacity (100%). The tracker configuration consistently charges faster and reaches 100% SoC earlier in the day compared to the fixed system (dashed green line). Once the battery is fully charged, the EMS has no choice but to curtail the remaining PV generation by exporting it to the grid. This confirms that during peak summer weeks, the BESS capacity is the primary bottleneck for self-consumption, and the tracker's additional yield is largely valorized through grid injection rather than local storage.

Figure 4 highlights the system's resilience during a harsh winter week sample (January 10-17), characterized by low irradiance and high heating loads.

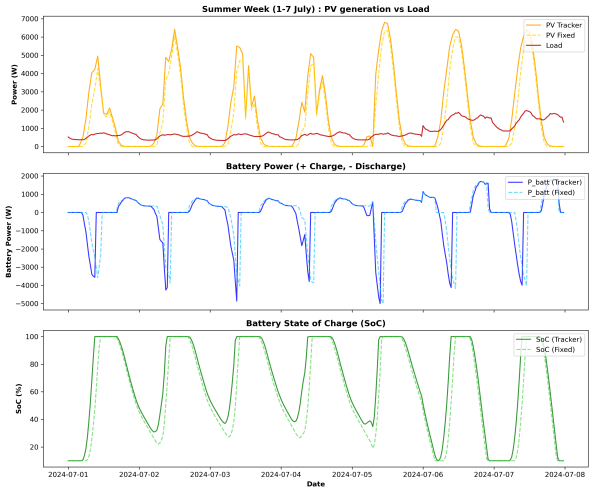


Figure 3: Summer week sample (July 1-7): PV generation vs Load (top), Battery power (middle), and SoC (bottom).

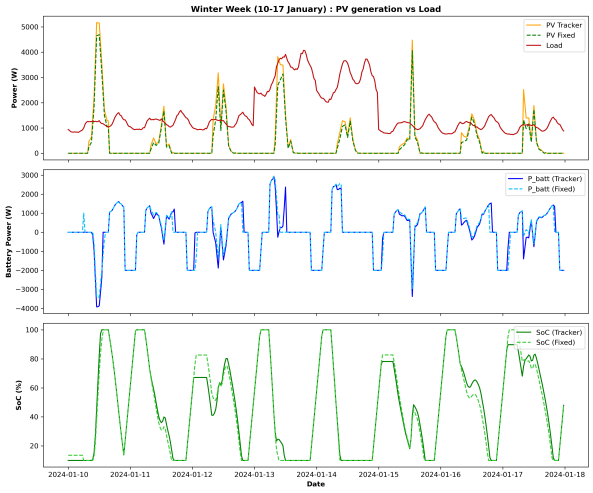


Figure 4: Winter week sample (January 10-17):PV generation vs Load (top), Battery power (middle), and SoC (bottom)

The top panel reveals that the PV production is Intermittent and often insufficient to meet the load, which frequently exceeds 3,000 W. Even with the tracker's optimized orientation, the PV contribution is marginal. The middle panel shows the EMS compensating for this deficit. The BESS undergoes deep, strong discharge cycles (positive power) during the evening hours, reaching up to +3,000 W to prevent expensive peak-hour grid imports.

The bottom panel (SoC) is particularly revealing: the battery frequently hits the lower critical threshold of 10% (e.g., around January 13th and 16th). During these periods, the SoC remains flat at 10% for several hours, indicating that the BESS is completely depleted and the grid must supply the totality of the household demand. Furthermore, the visible flat charging plateaus during the night (around -2,000 W) correspond to the "Case 4" (Off-Peak) EMS logic, where the system recharges from the grid to ensure enough capacity is available for the next day's morning and evening peaks. This confirms that in winter, the microgrid's autonomy is heavily constrained by the physical limits of the battery and the solar resource.

6.1.3. EMS Strategy Assessment: Peak Shaving and Window Shifting Efficiency

In accordance with the EMS architecture defined in Section 4, the data illustrate an optimized management of Peak (HP) and Off-Peak (HC) hours. To synthesize the intraday dynamics, Figure 5 displays the average 24-hour profiles for the summer (June-August) and winter (December-February) seasons.

During the summer period, the average daily PV generation drastically exceeds the household load. The tracker configuration yields a broader and higher power curve, peaking near 5,000 W. The EMS immediately exploits this surplus to charge the BESS. A critical observation comparing the two configurations is the temporal shift induced by the tracker. Due to its higher and earlier morning yield, the tracker's battery charging phase begins earlier than that of the fixed system. However, because the BESS has a finite energy capacity (10 kWh), this early charging inevitably leads to an earlier discharge phase (around 14:00 for the tracker and 15:00 for the fixed system). Consequently, the State of Charge (SoC) profile perfectly mirrors this dynamic: the tracker's SoC rises earlier, hits the 100% saturation ceiling sooner, and begins its afternoon decline earlier. This demonstrates that while the tracker yields more energy, the battery capacity becomes the primary bottleneck, forcing earlier grid exports during midday saturation.

Conversely, the winter average depicts a significantly harsher operating regime. The solar dynamics change drastically: the peak PV generation is roughly halved compared to summer, and the effective solar generation window shrinks significantly, spanning only from approximately 07:00 to 16:00 (compared to 05:00–18:00 in summer). Simultaneously, the household load increases substantially due to thermal demands (heating). The average daily load peaks at around 1,500 W during the day and up to 1,700 W in the evening, whereas summer loads rarely exceed 1,000 W.

In this constrained environment, the EMS shifts the BESS into a highly defensive, peak-shaving role, with

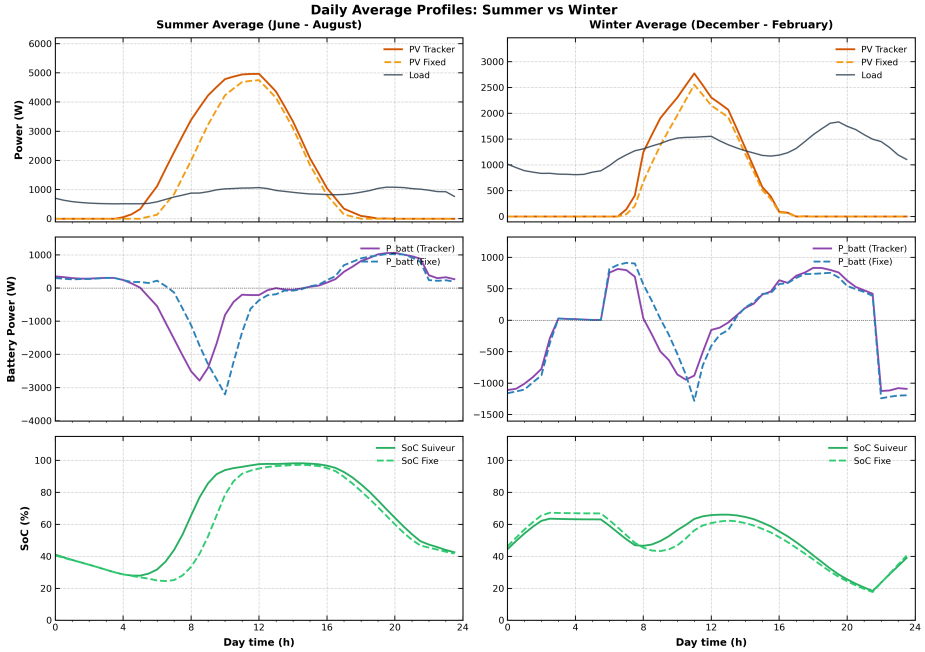


Figure 5: Average daily profiles for summer and winter: PV generation vs. Load (top), Battery power (middle), and SoC (bottom).

the battery undergoing deep discharge cycles to meet the elevated evening loads. The average winter profiles reveal specific EMS behaviors that illustrate this strategic adaptation. Around 06:00, the EMS initiates a charging phase (or maintains the existing charge) because the early household wake-up occurs while it is still dark, and no PV generation is available yet. Subsequently, during the day, the solar charging is considerably weaker than in summer, barely reaching $-1,000$ W (compared to $-3,000$ W in summer) due to the reduced irradiance. Finally, a strong recharge from the grid occurs during the night, visible as a large negative battery power. This nighttime grid charge is a deliberate EMS anticipation of the next day, which offers little solar production, ensuring the battery is sufficiently full to handle both the morning wake-up and the evening peak loads.

Comparing the tracker to the fixed system in winter reveals a notable advantage for battery health. The tracker effectively reduces the peak charging and discharging rates (the curves are flatter and less jagged). By spreading the reduced winter solar yield over a slightly longer window, the tracker smooths out the power transients, reducing the stress on the $LiFePO_4$ cells. Furthermore, the average winter SoC rarely exceeds 65% during the day, in stark contrast to summer's 100% saturation, and frequently plummets below 20% during the night. This confirms that during winter, the microgrid operates near its autonomy limits, heavily relying on EMS arbitrage (off-peak grid charging) and peak shaving to remain economically viable.

Ultimately, Figure 5 visually validates the strategic value of the EMS/BESS integration through Peak Shaving and Window Shifting mechanisms. In both seasons, the EMS explicitly targets the evening load peak (18:00–20:00). By discharging the battery during these expensive Peak tariff windows, it physically reduces the site's maximum demand drawn from the grid, flattens the consumption profile, and neutralizes tariff spikes. The EMS effectively decouples generation from consumption. In summer, it shifts surplus solar energy from midday to the evening. In winter, it shifts cheap off-peak grid energy (night) to cover the morning wake-up and the expensive evening peak. This temporal decoupling transforms the microgrid from a passive consumer into an active, flexible node, maintaining resiliency even when solar production is at its lowest. The averaged profiles highlight the high efficiency of the EMS algorithm. The system anticipates thermal loads and solar variance, maintaining the BESS strictly within its safe operational boundaries ($SoC \in [10\%, 100\%]$) while dynamically arbitraging between PV, load, and grid. The visual contrast between the summer (rapid saturation, shallow cycling) and winter (deep cycling, grid reliance) confirms that the EMS adapts its dispatch strategy to the

seasonal thermodynamic reality of Mulhouse.

6.2. Economic and Financial Viability Assessment

This section evaluates the economic feasibility of the proposed domestic microgrid in Mulhouse for the year 2024. The assessment relies on a 20-year projection model, incorporating detailed CAPEX, OPEX, and current French electricity tariffs.

6.3. Input Parameters and Financial Assumptions

The financial model is built upon specific tariffs, investment costs, and operational parameters defined for a residential microgrid. Table 6 summarizes the economic and technical assumptions used in the simulation.

Table 6: Economic and Financial Modeling Parameters

Parameter	**Value**
Tariffs and Grid Costs (EDF 2024)	
Peak Hours Tariff (C_{HP})	0.2700 €/kWh (06:00–22:00)
Off-Peak Hours Tariff (C_{HC})	0.2068 €/kWh (22:00–06:00)
Surplus Injection Tariff (C_{INJ})	0.1301 €/kWh (EDF OA)
Annual Grid Subscription (C_{FIXE})	170 €/year
Financial and Technical Assumptions	
Electricity Price Inflation (r_e)	3.0%
Discount Rate (d)	4.0%
PV Degradation Rate (δ)	0.5%/year
Inverter Replacement (Year 10)	1,000 €

The physical system consists of 36 m^2 of PV panels (20 modules of 380 Wc, totaling 7.6 kWc), coupled with a 10 kWh Huawei LUNA2000 BESS and an 8 kW hybrid inverter. Table 7 details the capital and operational expenditures for both scenarios.

Table 7: Detailed CAPEX and OPEX Breakdown for the Microgrid Systems

Component / Parameter	**PV Fixed + BESS**	**PV Tracker + BESS**
Capital Expenditures (CAPEX)		
PV Modules (20 x 120 €)	2,400 €	2,400 €
Hybrid Inverter (8 kW)	1,900 €	1,900 €
BESS (LUNA2000-10-S0)	5,800 €	5,800 €
Cables & Protection	800 €	800 €
IoT Controller & Sensors	300 €	300 €
Mounting Structure	700 €	700 €
Tracker Mechanical Surcharge	0 €	3,200 €
CAPEX	11,900 €	15,100 €
Operational Expenditures (OPEX)		
Annual Maintenance	120 €/year	180 €/year

6.4. 2024 Cash Flow Analysis: Monthly Dynamics

Figure 6 illustrates the dynamic monthly balance between grid import expenses and injection revenues for the year 2024 in Mulhouse.

The system exhibits a pronounced seasonal contrast, acting primarily as an energy consumer during winter and an energy producer during summer. During the winter months (November to February), solar irradiance is at its lowest while domestic electricity demand is at its peak, forcing the microgrid to remain heavily reliant on the grid. January represents the worst-case scenario, with grid import costs reaching 179.3 € for the Fixed system and 169.1 € for the Tracker, whereas injection revenue during these months is negligible. Furthermore, the 10 kWh BESS capacity is insufficient to bridge this seasonal winter deficit; it successfully performs daily peak-shaving but cannot provide seasonal energy shifting, meaning the grid remains the primary backup during winter.

From April to September, the dynamics reverse drastically as the microgrid shifts to a net exporter, generating substantial revenue through the EDF OA surplus injection tariff. The highest revenues are recorded in June and July; in June alone, the Fixed system generates +56.7 €, while the Tracker reaches +89.7 €. During these summer months, grid import costs also drop significantly thanks to both the high PV yield and the BESS covering evening consumption.

The monthly cash flow graph clearly illustrates the economic superiority of the dual-axis tracking system during high-irradiance periods. The Tracker consistently outperforms the Fixed system in injection revenue; for instance, in July, the Tracker generates +76.7 € compared to +49.3 € for the Fixed system, representing a +55% increase in monthly export revenue. Interestingly, the financial gap between the two systems narrows most significantly during the winter months, specifically in December where the difference is only 6.6 € (−167.9 € for the Fixed system and −161.3 € for the Tracker). This occurs because the Tracker's mechanical advantage relies heavily on Direct Normal Irradiance (DNI). During overcast winter days, diffuse solar radiation dominates, rendering the tracking mechanism ineffective and yielding similar low production for both systems.

Despite this winter convergence, the annual statistics presented in Table 5 clearly validate the Tracker's superiority over the full year. The Tracker produces +46.0% for grid sales than the Fixed system, and reduces annual grid purchases by 9.3%. This dynamic is perfectly captured by the self-sufficiency index (Tx_{auto}). The Fixed system achieves 56.2% self-sufficiency, meaning it covers just over half of the household's total energy needs.

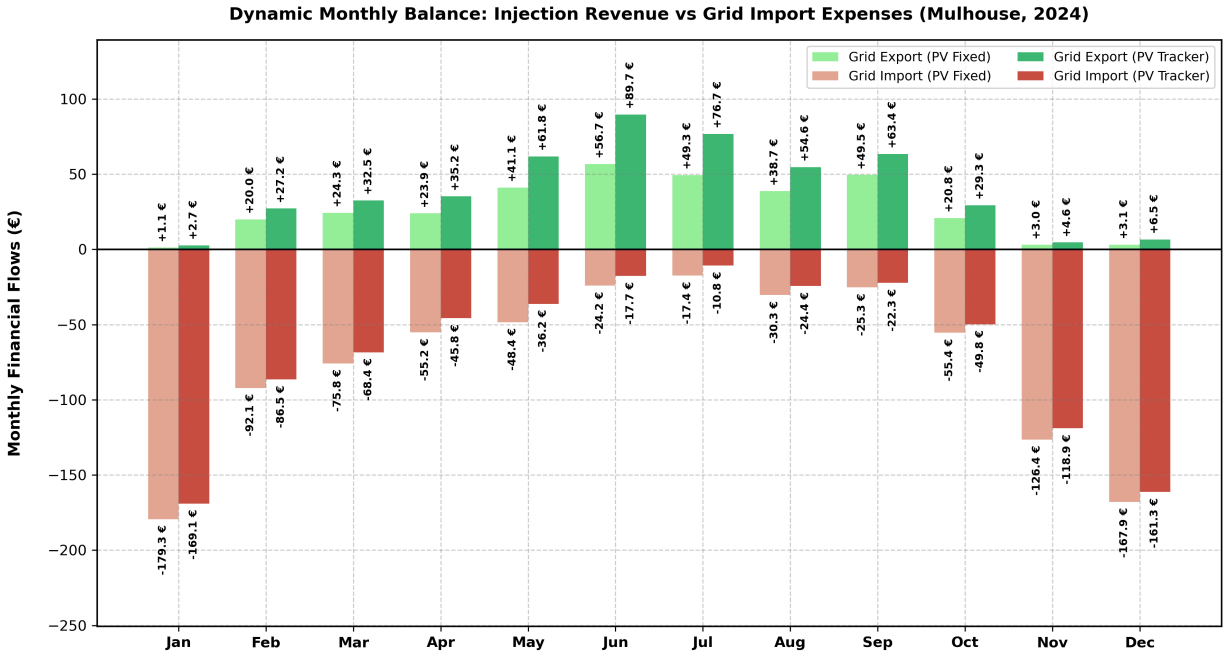


Figure 6: Dynamic Monthly Balance: Injection Revenue vs Grid Import Expenses (Mulhouse, 2024).

The Tracker reaches 60.2%, a gap of +4.1 percentage points. In practical terms, this 4.1 point gap signifies that the household relies significantly less on the grid for its daily operations, offering better protection against future electricity price inflation.

Finally, the monthly analysis reveals that the "break-even" periods,where injection revenue equals import costs, occur around April and October. This indicates that while the system is highly profitable in summer, the winter deficit must be carefully budgeted.

6.4.1. Financial Key Performance Indicators (KPIs)

The financial viability of any energy project cannot be assessed solely through physical performance; it requires a rigorous evaluation using Key Performance Indicators (KPIs). These indicators play a pivotal role in financial evaluation, allowing investors to quantify profitability, assess risk, and make secure, data-driven decisions regarding capital allocation.

The Net Present Value (NPV) is the total value generated by the project over its lifetime, adjusted for the time value of money. The Fixed system yields an NPV of 16, 023.50 €, while the Tracker achieves 16, 207.92 €. Although the Tracker has a higher initial CAPEX (15, 100 €vs. 11, 900 €), its significantly higher energy yield allows it to generate a marginally higher NPV. However, the difference between the two systems is only about 184 €over 20 years. Such a narrow margin makes it difficult to definitively declare one system superior based on NPV alone; additional KPIs are necessary to fully understand the financial trade-offs.

The Internal Rate of Return (IRR) provides a different perspective by calculating the annualized effective compounded return rate that makes the NPV of all cash flows equal to zero. The Fixed system offers an IRR of 14.80%, while the Tracker offers 12.87%. Both values comfortably exceed the chosen discount rate of 4.0%, confirming the overall profitability of the investment. However, the Fixed system's higher IRR is highly attractive for an investor seeking rapid and efficient returns on their initial capital. A higher IRR indicates that the project generates cash flows more quickly relative to its upfront cost, which is particularly relevant given the Tracker's substantial mechanical surcharge.

This dynamic is further highlighted by the Payback Period (PBP). The Fixed system recovers its initial investment in 8 years, whereas the Tracker requires 10 years to break even. The two-year delay for the Tracker is directly attributable to its higher CAPEX (3, 200 € for the tracking mechanism). While a longer payback period increases financial exposure to long-term uncertainties, both systems fall well within the 20-year operational lifespan, making them both viable long-term investments. The 20-year financial trajectory, presented in Figure 7,

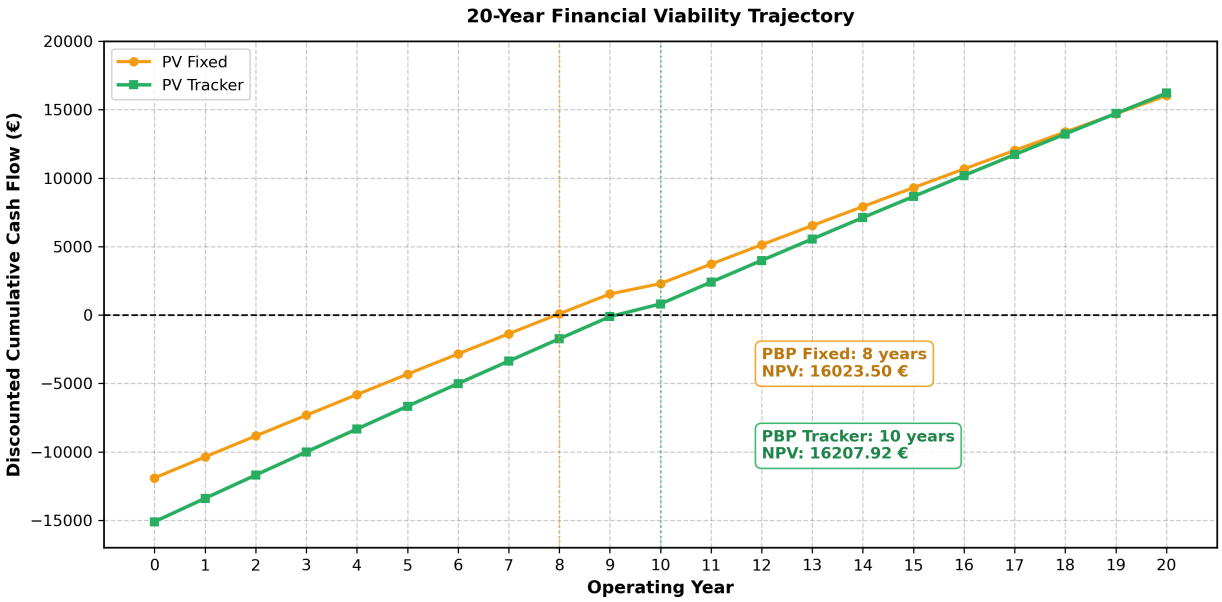


Figure 7: 20-Year Financial Viability Trajectory: Discounted Cumulative Cash Flow.

demonstrates the discounted cumulative cash flow for both configurations.

Finally, the Levelized Cost of Energy (LCOE) serves as a more sophisticated metric, as it evaluates the true, lifetime cost of generating one kWh of electricity. It integrates the initial CAPEX, annual OPEX, replacement costs (inverter at Year 10), the discount rate, and crucially, the 0.5% annual degradation of the PV cells. The Fixed system achieves an LCOE of 0.1352 €/kWh, compared to 0.1451 €/kWh for the Tracker. The Tracker's higher LCOE is a direct consequence of its heavier capital cost, which is not entirely offset by its increased energy production. Nevertheless, both values are significantly lower than the reference grid purchase price (averaging around 0.238 €/kWh).

While the Tracker slightly edges out the Fixed system in terms of absolute NPV due to higher total energy yield, the Fixed system emerges as the more financially conservative and attractive choice. It offers a faster payback (8 years), a higher return on investment (14.80% IRR), and a lower cost of energy (0.1352 €/kWh).

6.4.2. Sensitivity and Scenario Analysis

To evaluate the robustness of the investment against market volatility and regulatory uncertainties, a comprehensive sensitivity and scenario analysis was conducted on the PV Fixed system. The primary risk factors analyzed include the electricity price inflation rate (r_e), the discount rate (d), and variations in the initial Capital Expenditure (CAPEX). Table 8 summarizes the impact of these variables on the Net Present Value (NPV), the Internal Rate of Return (IRR), and the Levelized Cost of Energy (LCOE).

The extreme scenarios demonstrate the financial boundaries of the project. In the Best Case, a highly favorable environment combining a 10% CAPEX reduction (10,710 €), a surge in electricity prices to 5.50%, and a lower discount rate of 3.00% yields an exceptional NPV of 29,116.28 € and an IRR of 19.16%. Conversely, the Worst Case assumes a pessimistic reality with a 20% CAPEX overrun (14,280 €), a prolonged stagnation of energy prices at 1.00%, and a high discount rate of 6.00%. Even under these dire conditions, the project remains profitable, generating a positive NPV of 5,102.31 € and an IRR of 9.96%, thereby proving the inherent resilience of the investment.

Variations in initial costs directly influence the LCOE. A 20% decrease in CAPEX due to falling component prices (9,520 €) improves the IRR to 18.69% and lowers the LCOE to 0.1170 €/kWh. Conversely, a 20% increase in installation costs pushes the LCOE to 0.1641 €/kWh and the NPV down to 13,643.50 €, yet the project remains highly profitable. Regarding regulatory and financial risks, a government freeze on energy prices (simulated by a drop in inflation to 1.50%) reduces the NPV to 12,309.29 €. Similarly, an increase in the discount rate to 6.00% (simulating more expensive subsidized loans) penalizes discounted cash flows, lowering the NPV to

11,216.88 € and raising the LCOE to 0.1648 €/kWh.

The most pragmatic test is the Combined Scenario, which simulates hazardous market fluctuations with a conservative electricity inflation of 1.00% coupled with a 15% CAPEX overrun (13,685 €). Under this realistic stress test, the project still achieves a robust NPV of 9,420.12 € and an IRR of 10.57%, which comfortably exceeds the 4.00% discount rate. Ultimately, this analysis confirms the financial viability of the PV Fixed microgrid. Across all eight scenarios, the LCOE fluctuates between 0.1170 €/kWh and 0.1932 €/kWh, consistently remaining below the average retail grid price of approximately 0.238 €/kWh. This ensures long-term protection for the investor against market volatility, provided that the initial installation costs are strictly controlled.

Table 8: Scenario Analysis: Impact of Market Variations on Financial Viability (PV Fixed)

Scenario	r_e	CAPEX (€)	NPV (€)	IRR (%)	LCOE (€/kWh)
Base Case	3.00%	11,900	16,023.50	14.80%	0.1406
Best Case	5.50%	10,710	29,116.28	19.16%	0.1187
Worst Case	1.00%	14,280	5,102.31	9.96%	0.1932
Sens. CAPEX (+20%)	3.00%	14,280	13,643.50	12.04%	0.1641
Sens. CAPEX (-20%)	3.00%	9,520	18,403.50	18.69%	0.1170
Sens. r_e	1.50%	11,900	12,309.29	13.23%	0.1406
Sens. d	3.00%	11,900	11,216.88	14.80%	0.1648
Combined Sens.	1.00%	13,685	9,420.12	10.57%	0.1582

7. Conclusion

This study set out to evaluate the technical, energetic, and economic viability of a residential microgrid integrating dual-axis photovoltaic tracking, battery energy storage, and an advanced energy management system. The work builds upon a physical baseline of solar tracking optimization and expands it into a complete system-level framework for residential energy autonomy.

Globally, the deployment of residential PV-BESS systems is accelerating rapidly, particularly in Europe, North America, and Asia-Pacific, driven by falling battery costs and rising retail electricity tariffs. This international momentum underscores the relevance of the present study, which addresses the specific challenge of optimizing self-consumption under French tariff structures and semi-continental climatic conditions.

A key architectural contribution of this work is the geometric sizing of the inter-row spacing. For Mulhouse's latitude ($\phi = 47.75°$) and a critical winter tilt of 45°, the analytical model yielded a minimum center-to-center distance of $D \approx 8.00$ m. This constraint proved essential, as it allowed for the deployment of exactly two complete rows of modules on the available flat-roof footprint, resulting in a validated active area of $A_{pv} = 36.0$ m^2. while guaranteeing zero mutual shading losses during the critical winter solstice period.

On the photovoltaic modeling side, the integration of the Faiman thermal model was instrumental in capturing the temperature-dependent efficiency losses of the modules. By accounting for ambient temperature and wind-speed-driven convective cooling, the model enabled the calculation of an accurate annual average electrical conversion efficiency of $\eta_{annual} \approx 18.7\%$.

The dual-axis tracker was shown to increase annual energy yield by +19.52% compared to a fixed installation, rising from 8.05 to 9.63 MWh/yr, albeit with a marginally lower performance ratio due to slightly higher average cell temperatures.

The residential load modeling relied on the Enedis `RES11WE` profile for the Mulhouse climate zone (H1b, semi-continental). This dataset, processed at a 30-minute resolution, allowed for the reconstruction of a realistic annual demand of $9,000$ kWh. The scaling factor $\mathcal{F}_{scale} = 0.5158$ kW was determined to match this target, providing the instantaneous load power $P_{load}(t)$ essential for the EMS simulation. This step was crucial in capturing the structural mismatch between midday PV generation and morning/evening residential demand peaks.

The EMS and BESS integration demonstrated a transformative impact on the microgrid's operational performance. The rule-based dispatch algorithm, with its predictive off-peak charging and state-of-charge constraints, effectively performed peak shaving and tariff arbitrage. Compared to the fixed configuration, the tracker coupled with the EMS reduced grid purchases by −9.3% and increased grid sales by +46.0%. More importantly, the self-sufficiency rate improved from 56.2% to 60.2%, signifying a substantial reduction in grid dependency and enhanced protection against future electricity price volatility.

From a financial perspective, the Discounted Cash Flow analysis over a 20-year horizon confirmed the profitability of both configurations. The Net Present Value was slightly higher for the tracker, but its higher CAPEX diluted capital efficiency, resulting in a longer payback period and a lower Internal Rate of Return. The Levelized Cost of Energy (LCOE) was calculated at 0.1352 €/kWh for the fixed system and 0.1451 €/kWh for the tracker. Both values remain significantly below the average retail grid price of approximately 0.238 €/kWh, confirming the economic competitiveness of the investment.

The sensitivity analysis further validated the robustness of the investment. Across all eight scenarios tested, including worst-case assumptions of CAPEX overruns

and stagnant electricity prices, the project maintained a positive NPV and an IRR comfortably exceeding the discount rate, with the LCOE consistently remaining below the retail grid price. This confirms the financial resilience of the PV-BESS microgrid under market and regulatory uncertainties.

Future work should extend this framework in several promising directions. A deeper sensitivity analysis based on Monte Carlo simulation could provide probabilistic distributions of financial outcomes, offering a more nuanced risk assessment. The integration of an Electric Vehicle (EV) into the microgrid model, coupled with Vehicle-to-Grid (V2G) capabilities, would add a significant new dimension of flexibility and storage capacity. Thermal storage integration could further enhance the system's ability to shift heating loads. Finally, the IoT dimension, which was accounted for in this study through the inclusion of IoT sensors and controllers in the CAPEX, deserves dedicated modeling. A real-time dashboard, accessible via a mobile application, would empower the homeowner to monitor live PV production, consumption, battery state of charge, grid imports/exports, and the associated cash flows. Such a system could also include automated fault detection and alert mechanisms for PV or battery failures, ensuring proactive maintenance and sustained operational performance, thereby completing the vision of a truly smart and autonomous domestic microgrid.

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